From uncertain outcomes to a verifiable policy decision

Four jackets A–D · Success = high rating · Reference = 25% per jacket

Observational logs

Success-rate bounds

A	0.15 – 0.60
B	0.20 – 0.70
C	0.25 – 0.75
D	0.30 – 0.80

Randomized data

Transferred averages

$(m_A + m_C)/2 = .65$

$(m_B + m_D)/2 = .70$

One shared view

Keep all rates that satisfy both sources together.

Example allowed scenarios

A	B	C	D
0.55	0.60	0.75	0.80
0.60	0.70	0.70	0.70

Together these imply

$m_C \geq 0.70$

Reference mean = 0.675

Check a proposed policy

Lower bounds; target > 0.100 pp (IV-D)

Q1	Solve + check	**+1.000 pp**
Q2	Reuse + check	**+0.638 pp**
Q3	New solve + check	**+0.500 pp**

Find the best guaranteed gain

Always C guarantees		No larger guarantee
2.500 pp	=	**2.500 pp**

Matching checked bounds prove optimality

Constructed example · pp = percentage points

Every answer carries an independently checked proof